

Assignment # 1

Exploring Information Security Tools and Techniques

Due Date: October 31, 2025

Task 1: Password Hashing and Verification

Goal: Understand how hashing protects passwords.

1. Choose a password (e.g., "MySecurePass123").
2. Generate a SHA-256 hash using one of these methods:
 - o Windows:
 - o `certutil -hashfile password.txt SHA256`
 - o macOS / Linux:
 - o `echo -n "MySecurePass123" | shasum -a 256`
3. Change one character in the password and hash again.
4. Compare the two hashes.

Answer questions:

- What did you observe about how small changes affect the hash?
- Why are hashes important in password storage?

Task 2: File Encryption and Decryption

Goal: Learn to protect confidential data using encryption.

1. Create a text file `secret.txt` and write a short message.
2. Encrypt the file using one of these tools:
 - o 7-Zip: Right-click → Add to archive → Set password → Select "AES-256 encryption".
 - o GPG (Windows/macOS/Linux):
 - o `gpg --symmetric --cipher-algo AES256 secret.txt`
3. Try opening the encrypted file without the password.

Answer questions:

- What happens when you try to open the file without the key or password?
- Why is encryption important for data privacy?

Task 3: File Integrity Check

Goal: Verify whether a file has been changed.

1. Download any small file (e.g., a PDF or image).
2. Generate its MD5 or SHA256 hash and save it.
3. Open the file in an editor (add or change one word or pixel) and save again.
4. Generate the hash again and compare.

Answer questions:

- How does the hash change after modifying the file?
- Why is integrity checking used in software downloads?

Task 4: Network Traffic Exploration

Goal: Observe how data travels over a network.

1. Install Wireshark from <https://www.wireshark.org/>.
2. Run Wireshark and start capturing traffic for 1–2 minutes.
3. Stop the capture and observe:
 - o IP addresses communicating
 - o Common protocols (e.g., HTTP, DNS, TCP)
4. Take a screenshot of the Wireshark window (do not share private data).

Answer questions:

- What did you observe about network communication?
- Why should sensitive data (like passwords) be sent over encrypted connections?

Deliverables

Submit a single ZIP file containing and send to irfanuddin@awkum.edu.pk:

1. `password_hash.txt` — your hash results
2. `secret.txt.gpg` or `secret.7z` — your encrypted file
3. `file_hashes.txt` — before/after hash comparison
4. `answers.txt` — your short answers to each question
5. Screenshot of Wireshark capture
6. An `assignment.docx` file containing all the steps you have performed.